THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** second law of reflection	Statement of the second law.
2	**(c)** retina	Light-sensitive screen at the back.
3	**(a)** coulomb	1 C is the charge of about 6×10^{18} electrons.
4	**(c)** from the north pole to the south pole	By convention, outside the magnet.
5	**(c)** in vacuum to that in the medium	n = c/v.
6	**(c)** a concave lens	The concave lens diverges the rays.
7	**(c)** directly proportional to the potential difference, at constant temperature	V = IR.
8	**(a)** a bar magnet	One end behaves as N and the other as S.
9	**(d)** a converging lens	It converges parallel rays.
10	**(b)** dispersion	Newton's prism experiment.
11	**(a)** R1 + R2 + R3	Rs = R1 + R2 + R3.
12	**(d)** induced current in a coil due to a changing magnetic field	It was discovered by Faraday.
13	**(c)** virtual, erect and of the same size as the object	Plane mirror image is virtual, erect, same size, laterally inverted.
14	**(c)** outer surface of the cornea	The cornea has the greatest change in refractive index.
15	**(b)** ampere	1 A = 1 C/s.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Angle of incidence equals the angle of reflection (1); Incident ray, reflected ray and normal lie in the same plane (1).	1+1
17	Ability of the eye lens to adjust its focal length (1); To see both near and distant objects clearly (1).	1+1
18	Current when 1 coulomb of charge flows through a point in 1 second (2).	2
19	Right-hand thumb rule: hold the wire in the right hand with the thumb along the current (1); Curled fingers show the direction of the field (1).	1+1
20	Concave mirror: negative (1); Convex mirror: positive (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Virtual and erect (1); Same size as the object (1); Laterally inverted; as far behind as the object is in front (1).	1+1+1
22	Cornea: transparent front; most refraction (1); Iris and pupil control the light entering (1); Retina is the screen with rods and cones (1).	1+1+1
23	Q = I t = 0.5 x 600 = 300 C (1); n = Q / e (1); n = 300 / 1.6×10^{-19} = 1.875×10^{21} (1).	1+1+1
24	Lines emerge from N and merge at S outside the magnet (1); They form closed loops (1); Closer lines mean a stronger field and no two lines intersect (1).	1+1+1
	OR: 1/v = 1/f + 1/u = 1/10 - 1/30 (1); v = +15 cm (1); m = v/u = -0.5 (1).	1+1+1
25	Ciliary muscles weaken (1); The eye lens loses flexibility (1); The near point recedes (presbyopia) (1).	1+1+1
	OR: Rs = 2 + 3 + 5 = 10 ohm (1); I = V / R (1); I = 0.6 A (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	At infinity: at F, real, point-sized (1); Beyond C: between F and C, real, inverted, diminished (1); At C: at C, real, inverted, same size (1); Between C and F: beyond C, real, inverted, enlarged; at F: at infinity, enlarged (1); Between F and P: behind the mirror, virtual, erect, enlarged (1).	1+1+1+1+1
	OR: Cornea: transparent window that refracts light (1); Iris: controls pupil size (1); Pupil: allows light to enter (1); Lens: focuses the image on the retina (1); Retina and optic nerve: convert light into electrical signals sent to the brain (1).	1+1+1+1+1
27	V is proportional to I at constant temperature (1); Connect cell, ammeter, resistor and key in series and a voltmeter across the resistor (1); Take readings for different numbers of cells (1); The V-I graph is a straight line through the origin (1); The slope of the graph gives the resistance (1).	1+1+1+1+1
	OR: Place a compass near a straight wire passing through a card (1); Switch on the current: the needle deflects (1); Reversing the current reverses the deflection (1); Iron filings form concentric circles (1); Right-hand thumb rule gives the direction (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** the south pole to the north pole	They form closed loops.
2	**(b)** bends towards the normal	Glass is optically denser.
3	**(b)** a convex lens	A convex lens converges the rays.
4	**(d)** ohm	1 ohm = 1 V/A.
5	**(a)** makes an electromagnet	The core becomes magnetised.
6	**(c)** +2 D	P = 1/0.5 m = 2 D.
7	**(a)** Red	Red has the longest wavelength.
8	**(c)** current	Same current.
9	**(a)** Michael Faraday	Faraday, 1831.
10	**(d)** pole	The pole lies on the mirror.
11	**(b)** iris	Iris adjusts the amount of light.
12	**(b)** ammeter connected in series	An ammeter has a very low resistance.
13	**(d)** never intersect each other	Two directions at a point are impossible.
14	**(b)** bends away from the normal	Speed increases.
15	**(d)** in front of the retina	Eyeball is too long or lens too powerful.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Excessive curvature of the eye lens (1); Elongation of the eyeball (1).	1+1
17	Length and area of cross-section (1); Nature of the material and temperature (1).	1+1
18	Fleming's left-hand rule (1); Forefinger: field; middle finger: current; thumb: force (1).	1+1
19	The perpendicular distance between the original ray and the emergent ray (1); Seen in a rectangular glass slab (1).	1+1
20	Scattering of light by colloidal particles (1); Makes the path of light visible (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	A force acts on it (1); The force is perpendicular to the velocity and field (1); Its path becomes curved (1).	1+1+1
22	Speed decreases (1); Wavelength decreases (1); Frequency remains the same (1).	1+1+1
23	They are much closer to us (1); They are extended sources made of many point sources (1); The variations from them average out (1).	1+1+1
24	Filament has very high resistance, so heat produced is large (1); Heat H = I^2 R t (1); Connecting wires have low resistance (1).	1+1+1
	OR: Live: red (1); Neutral: black (1); Earth: green (1).	1+1+1
25	1/v = 1/f - 1/u = -1/15 + 1/30 = -1/30 (1); v = -30 cm (in front of the mirror) (1); Real, inverted and same size (m = -1) (1).	1+1+1
	OR: For near objects they contract and the lens becomes thicker (1); Focal length decreases (1); For distant objects they relax and the lens becomes thinner (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Power = 1/f in metres; unit dioptre (1); P1 = +4 D (1); P2 = -2 D (1); Net power = +2 D (1); Combination is converging with f = 50 cm (1).	1+1+1+1+1

	OR: Dispersion is splitting of white light (1); Light passes through a prism (1); Seven colour spectrum is formed (1); A second inverted prism recombines the colours (1); White light emerges, showing the prism does not create colours (1).	1+1+1+1+1
27	Work done to move charge Q across V is W = V Q (1); Q = I t (1); W = V I t (1); Using V = I R, W = I^2 R t (1); This energy is dissipated as heat H (1).	1+1+1+1+1
	OR: Mains supply of 220 V comes through live (red) and neutral (black) (1); Earth wire (green) is connected to the metal body (1); Fuse is placed in the live wire (1); Appliances are connected in parallel (1); Earth wire avoids shocks and fuse avoids overloading (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** a straight line through the origin	Slope = R.
2	**(c)** soft iron wrapped with a coil carrying current	It is a temporary magnet.
3	**(a)** dioptre	1 D = 1 m^-1.
4	**(b)** Violet	Shortest wavelength.
5	**(d)** potential difference	Same voltage.
6	**(b)** a magnet is moved towards or away from it	Relative motion changes the flux.
7	**(c)** R/2	f = R/2.
8	**(d)** accommodation	Done by ciliary muscles.
9	**(d)** voltmeter connected in parallel	A voltmeter has a very high resistance.
10	**(c)** the closeness of the field lines	Crowded lines mean a strong field.
11	**(a)** 2×10^8 m/s	v = c/n.
12	**(d)** farther than 25 cm	A person cannot see nearby objects clearly.
13	**(b)** directly proportional to its length	R is proportional to L.
14	**(c)** increasing the current or the number of turns	More turns, more current.
15	**(a)** negative	Concave lens has negative focal length.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	P = VI (1); P = I^2 R = V^2 / R (1).	1+1
17	AC can be transmitted over long distances with less loss of energy (1); It is cheaper to generate and distribute AC on a large scale (1).	1+1
18	Rays parallel to the principal axis meet at a point after reflection (1); This point on the axis is the principal focus (1).	1+1
19	Iris (1); It changes the size of the pupil (1).	1+1
20	Work done to move a unit charge from one point to another (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	W = V Q (1); W = 12 x 2 (1); W = 24 J (1).	1+1+1
22	A compass needle is placed near a wire (1); When current flows, the needle is deflected (1); Electric current produces a magnetic field (1).	1+1+1
23	1/v = 1/f + 1/u = -1/20 - 1/20 (1); 1/v = -1/10 (1); v = -10 cm (virtual image on the same side) (1).	1+1+1
24	Clouding of the lens of the eye (1); Causes loss of vision (1); Treated by surgery to replace the lens (1).	1+1+1
	OR: 1/R = 1/20 + 1/30 + 1/60 (1); 1/R = 6/60 = 1/10 (1); R = 10 ohm (1).	1+1+1
25	Stretch the thumb, forefinger and middle finger mutually perpendicular (1); Forefinger: field; middle finger: current (1); Thumb: force (1).	1+1+1
	OR: Negative m: real and inverted (1); Positive m: virtual and erect (1); Magnitude greater than 1 means enlarged (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Mirror formula 1/v + 1/u = 1/f (1); 1/v = -1/20 + 1/25 = -1/100 (1); v = -100 cm (1); m = -v/u = -4, so h' = -16 cm (1); Image real, inverted, magnified (1).	1+1+1+1+1
	OR: Accommodation is the adjustment of focal length (1); For near objects the lens becomes thicker (1); For far objects the lens becomes thinner (1); Near point 25 cm (1); Far point at infinity (1).	1+1+1+1+1

27	Same current I flows through each (1); V = V1 + V2 + V3 = I (R1 + R2 + R3) (1); Rs = R1 + R2 + R3 (1); Rs = 3 + 4 + 5 (1); Rs = 12 ohm (1).	1+1+1+1+1
	OR: Field is like that of a bar magnet (1); Uniform field inside, parallel to the axis (1); Soft iron core placed inside the solenoid (1); Core becomes strongly magnetised (1); Electromagnet is temporary: it works only while the current flows (1).	1+1+1+1+1

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